

0:00	Title	“Good morning. I’m Martin Maxa from the Department of Measurement, here at this faculty. This is joint work with Jakub Svatoš from the same department.” ■ today: detecting, classifying, localizing impulsive sounds – in our case, gunshots ■ main part: gunshot classification – cepstral features + two classifiers; localization briefly
0:45	Motivation	“This work is part of a larger project – an acoustic sensor system for protecting soft targets.” ■ good complement to cameras: omnidirectional, no line of sight ■ [CUT] full system: detect, classify (to calibre), localize – outdoors, noise, a lot of reflections ■ [CLICK] three questions: (1) which cepstral features best separate gunshots from similar impulsive sounds, (2) NN or SVM, (3) frame length vs. results
1:50	Signature	“Let’s start with the signal of a gunshot. A gunshot has one or two components, depending on the projectile speed.” ■ subsonic → muzzle blast only; supersonic → also a shock wave “this recording is a supersonic shot, so we can see both” ■ shock wave ≈ 2.5 ms – [CUT] along the bullet’s path, typical N-shape ■ [CLICK] muzzle blast ≈ 45 ms later – gases leaving the barrel, <i>every gunshot has it</i> ■ between and after = reflections – “this matters for the frame length, and I’ll come back to it in the results”
3:00	Single-domain	“This also shows why one domain alone is not enough.” ■ spectrum → tells you more about the environment than about the gunshot ■ waveform → changes as the sound travels and hits obstacles ■ ⇒ features combining both domains – cepstral features do exactly that
3:30	System	“Now to the system itself. It’s a set of independent sensor units connected to a remote server.” ■ unit monitors continuously → on an impulsive event, sends the signal ■ server does two things – classifies the event, localizes the source ■ more units: wider coverage + position of the source tested with several calibres and types of ammunition
4:20	Localize	“Let me briefly explain the localization first. Each sensor unit has two MEMS microphones, about twenty centimetres apart, sampled at 44.1 kilohertz.” ■ TDoA between the microphones → $\Theta = \arcsin(c \Delta n / (df_s))$ → direction to the source ■ [CLICK] one unit = one direction; two units → where they cross = position ■ 70 events, 9 reference angles – cross-corr + parabolic interpolation, MAE 0.77°
5:20	Features	“Now to the main part – classification. We compared four cepstral feature extractors. They differ in the filter bank they use.” ■ MFCC mel, dense low (from speech; works for gunshots too) LFCC linear IMFCC inverted mel, dense high GTCC gammatone, model of human hearing, more noise-robust (grid $4 \times 3 \times 2$ is on the slide) ■ NEXT (5:55) “And this is how the features are computed.” same chain: framing, FFT, filter bank, log, DCT filter bank = the only difference, all four on the same frequency axis
6:25	TABLE → CHART	“This is the full table of results for the nine-millimetre class.” ■ GTCC rows highlighted <i>don’t read the numbers</i> → NEXT (6:40) “Here are the same results, just the Matthews correlation coefficient.” ■ MCC: whole confusion matrix, less affected by class imbalance ■ red = GTCC on top in both panels, every frame length NN does better than SVM, gap grows with longer frames ■ best GTCC + NN, 50 ms → 99.15 / 96.77 / 0.98 ⇒ 117 of 118 correct (31 of them 9 mm; 30 of 31 recalled)
7:35	Discussion	“Three things to take from this.” ■ MFCC, LFCC, IMFCC similar – filter spacing makes little difference ■ GTCC clearly better – [CUT] hearing model, noise-robust: <i>likely</i> reason, not tested directly ■ NN slightly better than SVM, especially longer frames – <i>general trend, not every single result</i> ■ [CLICK] longer frames improve accuracy, <i>but</i> at 50 ms the frame already contains more and more reflections – “so even longer frames wouldn’t necessarily help”
8:45	Conclusion	“To sum up.” ■ distributed sensor system → detects, classifies, localizes impulsive sounds ■ cepstral features work well with both classifiers ■ GTCC best of the four; NN slightly better than SVM, especially with longer frames ■ all tested on real firearms, several calibres and types of ammunition
9:30		“Thank you.”

If asked, the numbers. Corpus 371 train / 118 test, five classes (false alarms 86/21, **9 mm 70/31**, 5.56 NATO SD 100/31, 7.62 Tokarev 55/17, .22 60/18) | 26 features per extractor, $\approx 75/25$ split | NN: MATLAB Neural Fitting, Levenberg–Marquardt, 370 iterations | bearing: cross-corr 0.84°, +parabolic **0.77°**, PBDE 3.24°, PBDE+linreg 3.91° | detection stage = median-filter trigger, RTSI 2019, down to SNR ≈ 5 dB | appendix = last 4 PDF pages: **A1** setup **A2** bearing **A3** other cepstral **A4** metrics (RLC = recall).

Running late? Three passages are marked **[CUT IF LATE]** in the notes – **Motivation** (the three functions), **Signature** (Mach cone), **Discussion** (probable reason GTCC wins). All three ≈ 35 s; dropping them breaks nothing. **Checkpoint**: start the table at **6:25**; past 6:55, cut all three.

Do not guess: 2D position error in metres, SNR limits of *this* classifier, firing distances in this dataset, provenance of the waveform on slide 3. “We did not characterize that in this study – Jakub Svatoš ran the campaign, I can put you in touch.”